

2.0 Problem-Solving Concepts

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2.1 Define problem-solving

In the simplest terms, problem solving means solving a problem. However, what is the problem, and how do we go about solving it?

A problem is a situation that requires unknown information to be found or a question to be answered. In mathematics, we often encounter them as word problems, in which a problem is presented as text rather than in mathematical notation.

Therefore, problem-solving is the process of identifying a problem, analyzing it, developing possible solutions, applying an appropriate solution, and checking the result.

When solving word problems, follow these general guidelines:

- Read the problem carefully and figure out what you are asked to find.
- Identify the unknown values and assign each of them a variable.
- Translate the words into an algebraic equation.
- Solve the equation using the rules of algebra.
- Answer the original problem.
- Check your result to ensure it makes sense.

2.2 Discuss types of problems

Well-Defined Problems

A well-defined problem has a clear starting point, a clear goal, enough information to work with, and rules or procedures that can be followed.

For example, a 1 200 MB file is transferred at 40 MB per second. How long will the transfer take?

Time = File Size / Transfer Rate

Time = $1\,200/40 = 30$ seconds

Ill-Defined Problems

An ill-defined problem does not have one obvious solution. Information may be missing, and different solutions may be reasonable.

For example, what is the best computer for a college student?

The answer depends on factors such as budget, required software, portability, storage, battery life, and operating system.

2.3 Differentiate between problem-solving with computers and general problem solving

General Problem-Solving

General problem-solving may use experience, judgment, observation, communication, research, calculations, and trial and error.

Problem-Solving With Computers

Computers are especially useful when a problem involves large amounts of data, repeated calculations, searching or sorting, or clearly defined rules that must be followed many times.

For example, a company must calculate overtime pay for 2 000 employees. A computer can apply the same rules quickly and consistently to all employees.

A computer does not automatically understand what the user intends. Appropriate data and clear instructions must be given.

Examples

Would a computer be particularly useful?

1. Calculating the average mark for 10 000 students.
2. Choosing which photograph looks best for a personal profile.
3. Deciding which applicant would be the best fit for a small project team.
4. Searching a database containing one million customer records.

2.6 Develop expressions and equations

To solve word problems, we must first know how to translate English phrases into algebraic expressions. Certain words provide us with cues as to which mathematical operations are to be used. These words appear frequently in phrases and we shall call them cue words.

<u>Phrases Describing Addition</u> Two more than a number The sum of a number and two A number increased by two Two is added to a number Two greater than a number A number plus two The total of two and a number	<u>Phrases Describing Subtraction</u> A number decreased by 5 Five less than a number. Five smaller than a number. Five fewer than a number. A number diminished by five. A number minus five. The difference between a number and five.
<u>Phrases Describing Multiplication</u> Double a number. Twice a number. The product of ____ and a number. Triple a number. Three times a number.	<u>Phrases Describing Division</u> A number divided by 6. One-sixth of a number. The quotient of a number and six.

When translating English phrases into algebraic expressions, we translate words which represent:

- unknown quantities into variables
- numbers and operations into their algebraic symbols.

Examples

Write an algebraic expression.

1. Four more than a number.
2. Triple a number.
3. Eight fewer than a number.
4. One-half more than a number.
5. The sum of four times a number and seven.
6. Three less than the quotient of a number and four.

Sometimes, more than one operation can be performed on an unknown at the same time. In this case, we must be extra careful to ensure that the order of operations is the same as was intended. This may require the use of parentheses to enforce that order.

7. A number is increased by 8 and then it is tripled.
8. Four times the sum of a number and ten.
9. Eight more than triple a number.
10. A number is increased by eight and then it is tripled.
11. One-third of the sum of a number and four.
12. Eight divided by the difference between a number and six.

Solving Word Problems Using Algebra

We will start with the simplest kind – straight number problems. In this type, write an expression as we did on the previous page, but now we add in cue words for “=” such as “is” or “is the same as”. This will allow us to write an equation that we can solve.

Examples

Four more than a number is 12. What is the number?

Eight fewer than a number is 32. What is the number?

Two-fifths of a number is 8. What is the number?

Six times a number is 15. What is the number?

Three more than 8 times a number is 99. What is the number?